Dynamics from Probability:
Predictive Operators and the Koopman Picture

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Abstract

The classical approach to dynamics, following Koopman, describes how observable functions evolve under a known transformation T — but this presupposes access to T. The prior question is: given a current observation Q and a future observation F, what does an observer predict about F given Q, without assuming the dynamics?

Making prediction precise forces an answer. The canonical answer is the regular conditional distribution of F given Q, a Markov kernel $\Pi_Q : \alpha \to \mathcal{P}(\beta)$ realising $P(F \in \cdot \mid Q = q)$ across all outcomes simultaneously. Its existence requires disintegrating a joint distribution — which requires a σ-additive probability measure on the observable algebra. That measure is not assumed here; it is the object constructed in [2] from coherent structured observations satisfying collective exhaustion.

Once Π_Q exists, composing with Q gives the map $Q_* : \Omega \to \mathcal{P}(\beta)$ sending each sample point to its conditional law. The central result — the predictive factorization theorem — shows that Q_* is the minimal sufficient statistic for future prediction. Indexing over time, the kernels $\{\Pi_t\}$ compose; their composition rule is the Chapman–Kolmogorov equation, derived from temporal coherence rather than assumed. From it emerge a semigroup of Markov operators $\{K_t\}$ and the duality $\int K_t g \, d\mu = \int g \, d(\mu \star \Pi_t)$. When the kernels are Dirac measures, the construction recovers the classical Koopman operator exactly.

The output is a triple $(\mathcal{P}(\beta), \nu_{Q_*}, \{K_t\})$ — predictive state space, stationary distribution, Markov semigroup. Setting $Q_t = h \circ T^t$ in a measure-preserving system identifies Q_* with the delay map $x \mapsto (h(T^n x))_{n \in \mathbb{Z}}$. The question the construction leaves open is when Q_* is injective modulo null sets — when the predictive state space faithfully represents the underlying state space X.

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The classical approach to dynamics, following Koopman [1], describes how observable functions evolve: the operator $U_T f = f \circ T$ acts by precomposing with a known transformation T. This presupposes access to T as a primitive. The question this paper asks is prior: given a current observation $Q : \Omega \to \alpha$ and a future observation $F : \Omega \to \beta$, what does an observer predict about F given Q? The dynamics, if there are any, should follow from the answer — not precede it.

Making prediction precise immediately generates a requirement. The canonical answer is the regular conditional distribution of F given Q: the unique Markov kernel $\Pi_Q : \alpha \to \mathcal{P}(\beta)$ realising $P(F \in \cdot \mid Q = q)$ simultaneously across all outcomes. Its existence rests on the Rokhlin disintegration theorem, which requires a σ-additive probability measure on the observable σ-algebra. That measure is not an extra assumption here. It is what [2] constructs: a coherent family of finitely additive charges on a directed system of observable algebras extends to a genuine probability measure exactly when it satisfies collective exhaustion, the condition ruling out persistent mass on globally vanishing events. The predictive construction cannot begin until that measure exists; collective exhaustion is the condition under which it does.

Once Π_Q exists, composing with Q gives the map $Q_* := \Pi_Q(\cdot, -) \circ Q : \Omega \to \mathcal{P}(\beta)$, sending each sample point to its conditional law. The predictive factorization theorem shows that every conditional expectation of a function of F factors through Q_*: it is the minimal sufficient statistic for future prediction. Indexing over time, the kernels $\{\Pi_t\}$ compose; their composition rule is the Chapman–Kolmogorov equation, derived from temporal coherence rather than assumed. This yields a semigroup of Markov operators $\{K_t\}$ and the Koopman–Perron duality connecting them with the pushforward operators $\{\mu \mapsto \mu \star \Pi_t\}$. When the kernels are Dirac measures, the construction collapses to the classical Koopman picture.

The output is a triple $(\mathcal{P}(\beta), \nu_{Q_*}, \{K_t\})$. Setting $Q_t = h \circ T^t$ in a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $h \in L^\infty$ identifies Q_* with the delay map $\Phi_h(x) = (h(T^n x))_{n\in\mathbb{Z}}$. The construction then leaves open a natural question: when is Q_* injective modulo null sets — when does the predictive state space faithfully represent the underlying state space X? The final section takes this as its starting point.

The paper is self-contained. The connection to [2] is motivational; the formalism starts from a probability space $(\Omega, \mathcal{F}, P)$ and two measurable maps:

- $Q : \Omega \to \alpha$, where $(\alpha, \mathcal{A})$ is a measurable space;
- $F : \Omega \to \beta$, where $(\beta, \mathcal{B})$ is a standard Borel space.

The standard Borel hypothesis on β is used once: it is the hypothesis under which the Rokhlin disintegration theorem [3] guarantees the existence of a regular conditional distribution. All other results hold in greater generality.

We write $\mathcal{P}(\beta)$ for the space of probability measures on β, equipped with its canonical (Giry) σ-algebra. The joint pushforward $\rho := P \circ (Q, F)^{-1}$ on $\alpha \times \beta$ has marginals $\rho_\alpha = P \circ Q^{-1}$ and $\rho_\beta = P \circ F^{-1}$.

The regular conditional distribution of F given Q is the unique (up to $P \circ Q^{-1}$-null sets) Markov kernel $\Pi_Q : \alpha \to \mathcal{P}(\beta)$ satisfying

$$\int_A \Pi_Q(q, B)\, d(P \circ Q^{-1})(q) \;=\; P(Q \in A,\ F \in B)$$

for all measurable $A \subseteq \alpha$ and $B \subseteq \beta$.

Definition 1 (Predictive kernel)**.** The *predictive kernel* is Π_Q, the disintegration kernel of $\rho = P \circ (Q, F)^{-1}$ over its first marginal: $\Pi_Q(q, A) = P(F \in A \mid Q = q)$.

The kernel is covariant under measurable maps: if $\pi : \alpha \to \alpha'$ is measurable and $Q' = \pi \circ Q$, then $\Pi_Q(q, \cdot) = \Pi_{Q'}(\pi(q), \cdot)$ for $(P \circ Q^{-1})$-almost every q, by uniqueness of disintegration.

Composing Π_Q with Q gives the map sending each sample point to its conditional law.

Definition 2 (Minimal predictive state map)**.** The *minimal predictive state map* is

$$Q_* : \Omega \to \mathcal{P}(\beta), \qquad Q_*(\omega) \;=\; \Pi_Q(Q(\omega), \cdot).$$

Since Π_Q is a Markov kernel it is measurable, and Q_* inherits measurability by composition.

Theorem 3 (Predictive factorization)**.** *For every bounded measurable $g : \beta \to \mathbb{R}$:*

$$E[g(F) \mid \sigma(Q)] \;=\; \int g\, dQ_*(\cdot) \quad P\text{-a.e.}$$

Proof. Let $h(\omega) := \int g\, dQ_*(\omega)$. We verify the two defining properties.

Measurability. The map $q \mapsto \int g\, d\Pi_Q(q, \cdot)$ is strongly measurable; since $Q_* = \Pi_Q(Q(\cdot), \cdot)$, the map h is $\sigma(Q)$-measurable.

Set-integral identity. For $S = Q^{-1}(A)$:

$$\begin{aligned}\int_S h\,dP &= \int_A \int g\,d\Pi_Q(q,\cdot)\,d(P\circ Q^{-1})(q) \quad \text{(pushforward)}\\ &= \int_{A\times\beta} g(f)\,d\rho(q,f) \quad \text{(disintegration)}\\ &= \int_S g(F)\,dP.\end{aligned}$$

The conclusion follows from a.e. uniqueness of conditional expectation. □

Corollary 4 (Predictive sufficiency). $E[g(F) \mid \sigma(Q)] = E[g(F) \mid \sigma(Q_*)]$ *P-a.e.*

Proof. $\sigma(Q_*) \subseteq \sigma(Q)$, and the conditional expectation given $\sigma(Q)$ is already $\sigma(Q_*)$-measurable by the theorem. □

The operator $(K_{Q_*}g)(q_*) = \int g\,dq_*$ on $\mathcal{P}(\beta)$ recasts the theorem as: $E[g(F) \mid \sigma(Q)] = (K_{Q_*}g)\circ Q_*$ P-a.e.

1 Dynamics and duality

To introduce dynamics, index over a discrete time horizon. For each $t \in \mathbb{N}$, let $\Pi_t : \gamma \to \mathcal{P}(\gamma)$ be a Markov kernel on the state space γ, with $\Pi_0(q,\cdot) = \delta_q$. A family $\{\Pi_t\}$ satisfying the Chapman–Kolmogorov equation

$$\Pi_{t+s}(q,\cdot) \;=\; \int_\gamma \Pi_t(q',\cdot)\,d\Pi_s(q,q') \quad \text{(i.e., } \Pi_{t+s} = \Pi_t \circ_\kappa \Pi_s)$$

is a *Markov semigroup kernel.* The associated *Markov operators* are

$$(K_t g)(q) \;:=\; \int g\,d\Pi_t(q,\cdot), \qquad g:\gamma\to\mathbb{R}.$$

Each K_t is positive, unital, and contractive on bounded measurable functions, with $K_0 = \mathrm{id}$ (since $K_0 g(q) = \int g\,d\delta_q = g(q)$); positivity, unitality, and contractiveness are immediate from the Markov property.

Proposition 5 (Semigroup property). $K_{t+s}g = K_t(K_s g)$ *for all bounded measurable g.*

Proof. $(K_{t+s}g)(q) = \int g\,d(\Pi_t\circ_\kappa \Pi_s)(q,\cdot) = \int\big(\int g\,d\Pi_t(q',\cdot)\big)\,d\Pi_s(q,q') = \int (K_t g)(q')\,d\Pi_s(q,q') = (K_t(K_s g))(q)$, where the exchange of integrals is justified by Bochner–Fubini. □

The pushforward of a measure μ on γ through the kernel is $P_t^*\mu := \mu \star \Pi_t = \int \Pi_t(q,\cdot)\,d\mu(q)$.

Theorem 6 (Koopman–Perron duality). *For every bounded measurable g and finite measure μ:*

$$\int (K_t g)\,d\mu \;=\; \int g\,d(P_t^*\mu).$$

Proof. $\int g\,d(P_t^*\mu) = \int g\,d(\mu\star\Pi_t) = \int_\gamma \int g\,d\Pi_t(q,\cdot)\,d\mu(q) = \int (K_t g)\,d\mu$, by Bochner–Fubini. □

When $\Pi_t(q,\cdot) = \delta_{\varphi_t(q)}$ for a measurable map $\varphi_t : \gamma\to\gamma$, the stochastic picture collapses.

Proposition 7 (Deterministic limit). *If $\Pi_t(q,\cdot) = \delta_{\varphi_t(q)}$, then $K_t g = g\circ\varphi_t$ and $\varphi_{t+s} = \varphi_t\circ\varphi_s$.*

Proof. $(K_t g)(q) = \int g\,d\delta_{\varphi_t(q)} = g(\varphi_t(q))$. The flow law follows from $\delta_{\varphi_{t+s}(q)} = (\Pi_t\circ_\kappa\Pi_s)(q,\cdot) = \delta_{\varphi_t(\varphi_s(q))}$ and injectivity of the Dirac embedding. □

In the deterministic case, $K_t g = g \circ \varphi_t = U_T^t g$: the Markov operator specialises to the classical Koopman operator. The Koopman–Perron duality $\int K_t g \, d\mu = \int g \, d(P_t^* \mu)$ becomes the change-of-variables formula $\int g \circ T^t \, d\mu = \int g \, d(T_*^t \mu)$.

The full structure produced by the construction is:

Definition 8 (Observable dynamical system)**.** An *observable dynamical system* is a triple $(\mathcal{P}(\beta),\ \nu_{Q_*},\ \{K_t\}_{t\in\mathbb{N}})$ where $\mathcal{P}(\beta)$ is the predictive state space (the image of Q_*), $\nu_{Q_*} = P \circ Q_*^{-1}$ is the pushforward of P along Q_*, and $\{K_t\}$ is the Markov semigroup on $\mathcal{P}(\beta)$.

Setting $Q_t = h \circ T^t$ in a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $h \in L^\infty$ specialises the construction. Under this choice, $Q_*(\omega) = (h(T^n\omega))_{n\in\mathbb{Z}}$ is the delay map Φ_h, and the observable dynamical system is the triple $(\Phi_h(X), \mu \circ \Phi_h^{-1}, \{K_t\})$.

The question the construction leaves open is whether Φ_h is injective modulo μ-null sets: whether the predictive state space $\Phi_h(X)$ faithfully represents X. Injectivity modulo null sets is equivalent to the observable σ-algebra $\mathcal{O}_h = \sigma(\{h \circ T^n : n \in \mathbb{Z}\})$ equalling $\mathcal{B}(X)$ modulo μ. The Koopman–Perron duality translates this into a function-space question: $\mathcal{O}_h = \mathcal{B}(X)$ modulo μ if and only if the algebra generated by $\{h \circ T^n\}$ is dense in $L^2(X, \mu)$. The key is that the L^2-closure of a function algebra equals the L^2-space of the σ-algebra it generates — a density bridge between σ-algebra generation and L^2 approximation.

The cyclic vector condition $\overline{\mathrm{span}}\{U_T^n h : n \in \mathbb{Z}\} = L^2(X, \mu)$ is sufficient but strictly stronger: reconstruction requires only that the *algebra* generated by the delay orbit be L^2-dense, not the linear span. They coincide only when U_T has simple spectrum.

When reconstruction holds, $\Phi_h(X)$ is the full state space X up to measure-theoretic identification. The compact Stone space of [2] — introduced as a technical device for σ-additivity — is, under the reconstruction condition, identified with X itself. The construction begins with a compact completion of the sample space and ends by recognising it as the state space.

References

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